MIL-P-48093A (AR)
3 October 1979
SUPERSEDING
MIL-P-48093 (PA)
3 June 1976

MILITARY SPECIFICATION

PROJECTILE, 8 INCH, HE, M509E1
HARDWARE NON-METALLIC FOR

This specification is approved for use by US ARMY ARMAMENT RESEARCH and DEVELOPMENT COMMAND and is available for use by all departments and agencies of the department of defense.

1. SCOPE

1.1 This specification covers the hardware non-metallic for one type of projectile designated as Projectile, 8 Inch, HE, M509E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC: 1320

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9241400 - Hardware, Non-Metallic

3. REQUIREMENTS

3.1 Material - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts - The parts shall comply with all requirements specified on Drawing (Dwg.) 9241400, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Workmanship - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.4 First Article Inspection - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspection - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

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4.3.1 Submission - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Spacer	9210125-1	25
Spacer	9210125-2	25
Pad, Forward Plate	9210126	25
Spacer, Large	9331765-1	25
Spacer, Large	9331765-2	25
Spacer, Small	9331766-1	25
Spacer, Small	9331766-2	25

4.3.2 Inspections to be Performed - See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection - See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48093A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9241400
	Hardware, Non-Metallic				NEXT HIGHER ASSEMBLY 9298200
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spacer (9210125-1&-2) Examination for defects	25 ea		3.2	4.4.2.1
	Pad, Forward Plate (9210126) Examination for defects	25		3.2	4.4.2.2
	Spacer, Large (9331765-1&-2) Examination for defects	25		3.2	4.4.2.3
	Spacer, Small (9331766-1&-2) Examination for defects	25		3.2	4.4.2.4

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078.

a. Sampling Plans - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48093A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Spacer	1 OF 1	9210125-1 and-2
			NEXT HIGHER ASSEMBLY: 9241400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter improper	See	.40%	3.2	Gage
102	Dimension "A" improper	4.5.2	.40%	3.2	Gage
Minor					
201	Radius not present all around (both ends)		.65%	3.2	Visual
202	Evidence of poor workmanship		.65%	3.3	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48093A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Pad, Forward Plate		9210126
			NEXT HIGHER ASSEMBLY 9241400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Outside diameter, max.	See	.65%	3.2	Gage
202	Evidence of poor workmanship	4.5.1	.65%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Spacer, Large		9331765
			NEXT HIGHER ASSEMBLY 9241400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile outside of tolerance zone		.40%	3.2	Gage
102	Dimension "A" improper	See	.40%	3.2	Gage
103	Thickness of rib improper	4.5.2	.40%	3.2	Gage
104	Width of rib improper		.40%	3.2	Gage
Minor					
201	Surface variations outside permitted zone		.65%	3.2	Visual
202	Blends at ends of ribs not present		.65%	3.2	Visual
203	Excess gating		.65%	3.2	Visual (see note)
204	Evidence of poor workmanship		.65%	3.3	Visual

NOTES: When this characteristic is in doubt, gaging may be used to determine acceptability.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-48093A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Spacer, Small		9331766
			NEXT HIGHER ASSEMBLY
			9241400

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile outside of tolerance zone		.40%	3.2	Gage
102	Dimension "A" improper	See	.40%	3.2	Gage
103	Thickness of rib improper	4.5.2	.40%	3.2	Gage
104	Width of rib improper		.40%	3.2	Gage
Minor					
201	Surface variations outside permitted zone		.65%	3.2	Visual
202	Blends at ends of ribs not present		.65%	3.2	Visual
203	Excess gating		.65%	3.2	Visual(see note)
204	Evidence of poor workmanship		.65%	3.3	Visual

NOTES:
When this characteristic is in doubt, gaging may be used to determine acceptability.

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4.4.3 Testing - Testing is described in the First Article and Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test Methods and Procedures

4.5.1 Dimensional control of stamped parts - In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Dimensional Control of Molded or Plastic Parts - In place of the normal sampling and inspections associated with the Classification of Defects, and after a curing time for the parts has been approved (See Note 1) a sample of four (4) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for at least the defects listed in paragraphs 4.4.2.1 through 4.4.2.4 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the four (4) samples, one (1) sample shall be the last

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part produced. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defetive part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request. Contractor designs of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, ARRADCOM, Dover, New Jersey 07801, ATTN: DRDAR-QAR-I for approval prior to manufacture of equipment. The noted sub-paragraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1: In establishing a curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60 etc mins) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to ARRADCOM, Dover, New Jersey 07801 ATTN: DRDAR-QAR-Q. If there is a change in material, or in the cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C - The non-metallic hardware shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing

5.2.1 Level C - The non-metallic hardware preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with MIL-STD-1169. Each pack shall contain only one part number.

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5.3 Marking - Marking shall comply with MIL-STD-1169.

5.4 Shipping - When shipments from more than one lot are made at one time, such as in a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended Use - The components covered by this specification are intended for use on the Projectile, 8 Inch, HE, M509E1.

6.2 Ordering Data - See MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval - See MIL-A-48078. Submit equipment designs as required to Commander, ATTN: DRDAR-QAR-I, ARRADCOM, Dover, NJ 07801.

6.4 Submission of Results of Contractor-Conducted Examinations and Tests - Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1423 for the contract.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1320-A507

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